

CHOSHI, Japan

WMO: 476480

Lat: 35.7397N

Lon: 140.8572E

Elev: 92

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 32.8	(c) 34.6	(d) 13.6	(e) 11.0	(f) 39.4	(g) 16.2	(h) 12.5	(i) 39.6	(j) 40.7	(k) 45.0	(l) 34.6	(m) 46.2	(n) 12.5	(o) 290	(p) 0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.4	(c) 86.4	(d) 78.7	(e) 84.9	(f) 78.2	(g) 83.3	(h) 77.6	(i) 80.0	(j) 84.0	(k) 79.1	(l) 83.0	(m) 78.3	(n) 81.9	(o) 13.9	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 78.9	(b) 150.7	(c) 82.1	(d) 78.0	(e) 145.9	(f) 81.5	(g) 77.1	(h) 141.7	(i) 80.9	(j) 43.7	(k) 84.1	(l) 42.7	(m) 83.2	(n) 41.9	(o) 82.2	(p) 83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 33.2	(b) 28.8	(c) 25.6	(e) 29.5	(f) 90.3	(g) 1.8	(h) 2.0	(i) 28.2	(j) 91.8	(k) 27.2	(l) 92.9	(m) 26.2	(n) 94.1	(o) 24.9	(p) 95.5		
			DB	WB	25.1	81.2	1.7	1.2	23.8	82.1	22.8	82.7	21.8	83.4	20.5	84.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	43.5	44.0	49.2	56.8	63.7	68.6	75.0	78.7	74.4	66.7	57.4	48.4
	DBStd	12.74	4.25	4.65	5.45	5.05	4.31	3.89	4.22	3.69	4.37	4.21	4.77	5.02
	HDD50	570	208	177	83	8	0	0	0	0	0	0	3	91
	HDD65	2849	667	587	489	246	75	12	1	0	1	29	229	513
	CDD50	4453	6	10	59	214	423	558	776	889	731	519	226	42
	CDD65	1258	0	0	0	1	33	120	312	425	282	83	2	0
	CDH74	6896	0	0	0	0	18	181	1712	3426	1448	111	0	0
	CDH80	1289	0	0	0	0	0	5	293	770	218	3	0	0
(14) Wind	WSAvg	12.7	12.8	13.5	14.3	14.0	12.5	10.9	11.7	11.4	12.9	13.8	12.4	12.3
(15) Precipitation	PrecAvg	65.4	3.5	3.7	5.7	5.0	5.6	6.5	4.2	4.6	8.1	9.6	5.3	3.3
	PrecMax	92.8	10.7	9.2	14.5	9.7	14.1	13.0	12.3	13.9	18.4	27.9	15.5	11.7
	PrecMin	42.0	0.3	0.6	1.7	1.4	1.1	1.4	0.5	0.0	0.7	2.2	0.6	0.2
	PrecStd	10.1	2.4	2.0	2.3	1.9	2.6	2.6	2.6	3.4	4.0	5.5	3.2	2.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.5	62.6	65.6	70.7	76.2	80.6	87.2	89.1	86.5	79.9	71.1	66.9
		MCWB	55.2	57.7	58.0	63.1	67.3	73.8	78.4	79.5	78.2	73.7	64.9	62.3
	2%	DB	56.8	58.4	62.9	68.0	73.7	78.0	85.1	87.2	84.6	76.8	68.8	62.4
		MCWB	50.9	53.6	57.4	61.7	66.2	71.9	77.9	79.2	77.9	71.7	63.7	57.0
	5%	DB	53.7	55.0	60.7	66.1	71.9	76.0	83.4	85.8	82.7	74.7	67.0	59.3
		MCWB	47.8	49.4	55.7	60.4	65.3	70.7	77.2	78.8	77.1	69.7	61.8	53.4
	10%	DB	51.1	52.2	58.5	64.4	70.1	74.3	81.4	84.4	80.5	72.8	65.3	56.8
		MCWB	45.1	46.4	53.5	59.2	64.6	70.0	76.4	78.3	76.2	67.9	60.1	51.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.0	59.1	61.2	66.1	70.2	75.2	79.9	81.7	80.1	75.6	67.7	63.8
		MCDB	59.6	61.6	63.0	67.6	72.7	78.5	84.5	85.5	83.5	78.3	69.5	66.1
	2%	WB	52.8	54.9	58.7	63.9	68.6	73.5	78.8	80.6	79.0	73.4	65.2	58.3
		MCDB	55.6	57.5	61.8	66.7	71.4	76.2	83.1	84.7	82.5	76.0	67.6	61.3
	5%	WB	48.9	50.7	56.8	62.0	67.2	72.3	77.9	79.7	78.0	71.1	63.3	54.7
		MCDB	52.4	54.0	60.2	65.0	70.2	74.7	81.8	83.7	81.3	73.6	65.8	58.1
	10%	WB	45.9	47.5	54.0	60.1	65.8	71.0	77.0	78.9	77.0	69.2	61.3	52.1
		MCDB	50.3	51.2	57.7	63.4	68.9	73.3	80.4	82.8	80.1	71.8	64.2	56.0
(35) Mean Daily Temperature Range	5% DB	MDBR	11.9	11.2	10.4	9.9	9.1	8.0	8.6	8.4	7.2	7.5	10.1	11.5
		MCDBR	15.1	15.2	13.0	12.0	11.5	10.3	11.4	10.3	9.6	9.7	11.7	13.9
		MCWBR	12.5	13.3	11.7	8.7	6.7	5.6	5.5	4.4	5.0	7.5	9.9	11.8
	5% WB	MCDBR	13.8	13.8	12.0	10.6	10.4	9.2	10.6	9.9	9.2	8.7	10.6	12.7
		MCWBR	13.1	13.4	12.2	9.3	7.6	5.8	5.5	4.8	5.2	7.8	10.2	12.2
(40) Clear-Sky Solar Irradiance	taub		0.382	0.434	0.501	0.547	0.557	0.576	0.573	0.547	0.493	0.473	0.427	0.385
	taud		2.240	2.091	1.933	1.870	1.903	1.928	1.984	2.051	2.174	2.139	2.184	2.246
	Ebn at Noon		251	250	243	239	239	234	233	237	243	234	233	241
	Edh at Noon		34	44	56	63	62	60	57	52	44	42	36	32
(44) All-Sky Solar Radiation	RadAvg		822	967	1248	1480	1611	1459	1618	1646	1258	938	753	717
	RadStd		59	108	87	148	159	133	265	155	103	90	66	54

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page